Lesson 2: Mathematical Background

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2.1 Introduction

2.2 Points, Vectors and Notation

2.2.1 Sub topic

Recall the Cartesian plane, is defined by two perpendicular number lines: the xx-**axis**, which is horizontal, and the yy-**axis**, which is vertical. Using these axes, we can describe any **point** in the plane using an ordered pair of numbers. A **point** is a location in a given vector space. Therefore, **points** are Cartesian Coordinates e.g. p = [x, y]T or q = [x, y, z]T.
The Cartesian plane extends infinitely in all directions. To show this, math textbooks usually put arrows at the ends of the axes in their drawings.
Much of Computer Graphics involves discussion of points in 2D or 3D.
The superscript [...]T denotes transposition of a **vector,** so points p and q are column vectors (coordinates stacked on top of one another vertically). **Vectors** quantities have both direction and magnitude.

2.3 Basic Vector Algebra

Just as we can perform basic operations such as addition, multiplication etc. on scalar values, so we can generalize such operations to vectors. Figure 1.1 summarizes some of these operations in diagrammatic form.

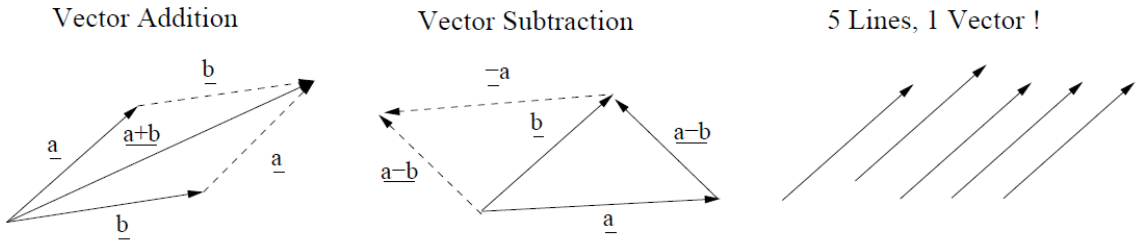


Figure 1.1: Illustrating vector addition (left) ,subtraction (middle). Right: lines (sometimes called 'rays') are vectors plus a starting point.

2.3.1. Vector Addition

When we add two vectors, we simply sum their elements at corresponding positions. So for a pair of 2D vectors a = [u, v]T and b = [s, t]T we have: a + b = [u + s, v + t]T

2.3.2. Vector Subtraction

Vector subtraction is identical to the addition operation with a sign change, since when we negate a vector we simply flip the sign on its elements.
−b = [−s,−t]T
a − b = a + (−b) = [u − s, v − t]T

2.3.3. Vector Scaling

If we wish to increase or reduce a vector quantity by a scale factor λ then we multiply each element in the vector by λ.

λa = [λu, λv]T

2.3.4. Vector Magnitude

We write the length of magnitude of a vector s as |s|. We use Pythagoras' theorem to compute the magnitude:

$$|\underline{a}| = \sqrt{u^2 + v^2}$$

This to be valid, since u and v are distances along the principal axes (x and y) of the space, and so the distance of vector **a** from the origin is the hypotenuse of a right-angled triangle. If we have an n-dimensional vector q = [q1, q2, q3, q..., qn] then the definition of vector magnitude generalizes to:

$$|\underline{q}| = \sqrt{q_1^2 + q_2^2 + q_{...}^2 + q_n^2} = \sqrt{\sum_{i=1}^{n} q_i^2}$$

2.3.5. Vector Normalization

We can normalize a vector **a** by scaling it by the reciprocal of its magnitude:

$$\hat{\underline{a}} = \frac{\underline{a}}{|\underline{a}|}$$

This produces a normalized vector pointing in the same direction as the original (un-normalized) vector, but with unit length (i.e. length of 1). We use the superscript 'hat' notation to indicate that a vector is normalized e.g. ˆa.

2.3.6. Vector Multiplication

We can define multiplication of a pair of vectors in two ways: the dot product (sometimes called the 'inner product', analogous to matrix multiplication), and the cross product (which is sometimes referred to by the unfortunately ambiguous term 'vector product').

2.3.6.1 Dot Product

The dot product sums the products of corresponding elements over a pair of vectors.

Given vectors a = [a1, a2, a3, a..., an]T and b = [b1, b2, b3, b..., bn]T , the dot product is defined as:

a ◦ b = a1b1 + a2b2 + a3b3 + ... + anbn

$$= \sum_{i=1}^{n} a_i b_i$$

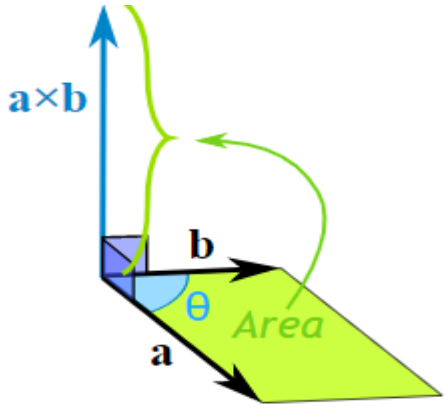
The dot product is both symmetric and positive definite. It gives us a scalar value that has three important uses in Computer Graphics and related fields. First, we can compute the square of the magnitude of a vector by taking the dot product of that vector and itself:

$$\begin{aligned} \underline{a} \circ \underline{a} &= a_1 a_1 + a_2 a_2 + ... + a_n a_n \\ &= \sum_{i=1}^{n} a_i^2 \\ &= |\underline{a}|^2 \end{aligned}$$

Second, we can more generally compute a ◦ b, the magnitude of one vector **a** in the direction of another **b**, i.e. projecting one vector onto another.
Third, we can use the dot product to compute the angle θ between two vectors (if we normalize them first). This relationship can be used to define the concept of an angle between vectors in n−dimensional spaces. It is also fundamental to most lighting calculations in Graphics, enabling us to determine the angle of a surface (normal) to a light source.
a ◦ **b** = |a||b|cosθ

2.3.6.2. Cross Product

The Cross Product **a** × **b** of two vectors is **another vector** that is at right angles to both. The magnitude (length) of the cross product equals the area of a parallelogram with vectors **a** and **b** for sides:



Taking the cross product (or "vector product") of two vectors returns us a vector orthogonal to those two vectors. Given two vectors a = [ax, ay, az]T and B = [bx, by, bz]T , the cross product a × b is defined as:

$$\underline{a} \times \underline{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}$$

In this course we only consider the definition of the cross product in 3D. An important Computer Graphics application of the cross product is to determine a vector that is orthogonal to its two inputs. This vector is said to be normal to those inputs, and is written **n** in the following relationship (care: note the normalization):
a × b = |a||b| sinzxzθˆ**n**

2.4 Matrix Algebra

A **matrix** is a rectangular array of numbers. Both **vectors** and **scalars** are degenerate forms of matrices. By convention we say that an (n×m) matrix has n rows and m columns; i.e. we write (height × width). In this subsection we will use two 2 × 2 matrices for our examples:

$$\underline{\underline{A}} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\underline{\underline{B}} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

2.4.1. Matrix Addition

Matrices can be added, if they are of the same size. This is achieved by summing the elements in one matrix with corresponding elements in the other matrix:

$c_{11}=a_{11}+b_{11}$

$c_{12}=a_{12}+b_{12}$

$c_{21}=a_{21}+b_{21}$

$c_{21}=a_{22}+b_{22}$

2.4.2. Matrix Scaling

Matrices can also be scaled by multiplying each element in the matrix by a scale factor.
Again, this is identical to vector scaling. Therefore, s**A** where s is scale factor and A is a matrix.

2.4.3 Matrix Multiplication

As we will see in Chapter 5, matrix multiplication is a cornerstone of many useful geometric transformations in Computer Graphics. You should ensure that you are familiar with this operation.

2.4.4 Matrix Inverse and the Identity

The identity matrix I is a special matrix that behaves like the number 1 when multiplying scalars (i.e. it has no numerical effect):

$IA = A$

The identity matrix leads us to a definition of the inverse of a matrix, which we write A^{-1}. The inverse of a matrix, when pre- or post-multiplied by its original matrix, gives the identity:

$AA^{-1} = A^{-1}A = I$

2.4.5 Matrix Transposition

Matrix transposition, just like vector transposition, is simply a matter of swapping the rows and columns of a matrix. As such, every matrix has a transpose. The transpose of **A** is written **A**T

Lesson 2 Discussion Session

1. Determine the cross product of a = (2,3,4) and b = (5,6,7)
2. You are required to determine the inverse of the matrices given below and also give the transpose form of it

a. $\mathbf{M}=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$

b. $\begin{bmatrix}1 & 0 & 0\\ 0 & 2 & 2\\ 3 & 0 & 8\end{bmatrix}$